

邹苏南

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个人简介

邹苏南是北京大学计算机学院计算机系统结构专业的四年级直博生，导师为罗国杰长聘副教授，隶属于高效计算与应用中心（CECA）的智能异构系统（DASYS）实验室。他于2022年获得北京大学信息科学与技术学院计算机科学与技术专业学士学位。当前研究兴趣主要包括大模型系统与基础设施、领域定制加速器编译框架和AI编译器；目前重点关注新型模型架构的算子-调度协同设计，以及模型压缩与适配算子/架构设计。他在电子设计自动化，尤其是逻辑综合优化方面也有丰富经验。其博士研究围绕Sketch算法的系统级应用、硬件优化与加速器自动生成展开。

教育背景

计算机系统结构专业 博士 北京大学计算机学院 导师：罗国杰教授	2022年9月 – 2027年6月（预计） 中国，北京
计算机科学与技术专业 学士 北京大学信息科学与技术学院	2018年9月 – 2022年7月 中国，北京

研究经历

小红书 Dots Ace 顶尖实习生 研究主题：LLM 中 xHC 算子融合与调度优化	2026年5月 - 至今 中国，北京
日本国立信息学研究所 NII 交换实习生，指导老师：Junichi Yamagishi 教授 研究主题：考虑演奏法的小提琴音频模型	2025年10月 - 2026年4月 日本，东京
卡耐基梅隆大学电子与计算机工程系 研究实习生，指导老师：Lawrence Pileggi 教授 研究主题：图神经网络指导 FPGA 逻辑锁隐藏设计。	2021年3月 - 2021年9月 远程
北京大学信息科学技术学院 研究实习生，指导老师：罗国杰教授 研究主题：强化学习辅助的逻辑优化算法。	2019年7月 - 2022年1月 中国，北京

发表论文

- [C9] **Sunan Zou**, Zhe Zhang, Yigitcan Özer, Guojie Luo, and Junichi Yamagishi. "VioLM: A Neural Language Model for Violin Synthesis with Articulation". *The 27th International Society for Music Information Retrieval Conference (ISMIR)*, 2026. (To Appear)
- [C8] Xiao Tan, Chenyue Li, **Sunan Zou**, and Guojie Luo. "RTLPilot: Skill-Driven Multi-Agent System for Repository-Level RTL Code Migration". *2026 International Symposium of EDA (ISED)*, 2026. (To Appear)
- [C7] **Sunan Zou**, Xueting Sun, Ziyun Zhang and Guojie Luo. "UltraSketchLLM: Sub-1-Bit LLM Compression via Sketch and Hardware-Friendly Operators". *63st Design Automation Conference (DAC)*, 2026. (To Appear)
- [C6] **Sunan Zou**, Bizhao Shi, Ziyun Zhang and Guojie Luo. "MuSA: Multi-Sketch Accelerator with Hybrid Parallelism and Coalesced Memory Organization". *IEEE 42nd International Conference on Computer Design (ICCD)*, 2024.

- [C5] Xinming Wei, Ziyun Zhang, **Sunan Zou**, Kaiwen Sun, Jiahao Zhang, Jiaxi Zhang, Ping Fan, and Guojie Luo. “AceRoute: Adaptive Compute-Efficient FPGA Routing with Pluggable Intra-Connection Bidirectional Exploration”. *43th IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2024.
- [C4] Bizhao Shi, Tuo Dai, **Sunan Zou**, Xinming Wei, and Guojie Luo. “ImageMap: Enabling Efficient Mapping from Image Processing DSL to CGRA”. *30th International European Conference on Parallel and Distributed Computing (Euro-Par)*, 2024.
- [C3] **Sunan Zou**, and Guojie Luo. “PONO: Power Optimization with Near Optimal SMT-based Sub-circuit Generation”. *61st Design Automation Conference (DAC)*, 2024.
- [C2] **Sunan Zou**, Jiaxi Zhang, and Guojie Luo. “Incremental SAT-based Exact Synthesis”, *34th Great Lakes Symposium on VLSI (GLSVLSI)* 2024. **[Best Paper Reward]**
- [C1] **Sunan Zou**, Jiaxi Zhang, Bizhao Shi, and Guojie Luo. “BESWAC: Boosting Exact Synthesis via Wiser SAT Solver Call”, *27th Design Automation and Test in Europe (DATE)*, 2024.
- [J2] Lei Chen, Yiqi Chen, Zhufei Chu, Wenji Fang, Tsung-Yi Ho, et al, and **Sunan Zou (Alphabetical Order)**. “Large circuit models: opportunities and challenges”. *Science China Information Sciences Volume 67*, 2024.
- [J1] **Sunan Zou**, Jiaxi Zhang, Bizhao Shi, and Guojie Luo. “PowerSyn: A logic synthesis framework with early power optimization”, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems Volume: 43 (TCAD)*, 2023.

代表性项目

[大模型训练系统] xHC 算子融合与训练调度优化

2026 年 5 月 – 至今

内容: 面向 xHC 在大模型训练中引入的细粒度计算与多 stream 状态传递开销，在 Megatron Core 中开展算子与调度协同优化。设计适配 xHC 张量形状与访存特征的计算内核，并基于计算依赖、数据复用和 GPU 执行特征选择融合边界；协同设计 Triton 前反向实现与自定义 Autograd 路径，使激活保存、重计算及稀疏状态物化策略与融合边界相匹配。调度层面探索跨子层共享控制计算，以及 CUDA Graph 和 combined 1F1B 下的状态管理；进一步探索在 PP/VPP 边界提前传输非活跃 stream，将状态通信与活跃路径计算重叠。通过数值对齐、性能分析与端到端训练验证正确性和优化效果。

角色: 主要贡献者。

[大语言模型算法] 少样本乐器音频生成语言模型构建

2025 年 10 月 – 2026 年 4 月

内容: 本项目聚焦于高质量少样本乐器音频生成模型的两阶段迁移训练框架和数据生成范式设计。首先，本项目设计了考虑演奏法的乐谱词元化方法，并依此构建乐谱合法化-力度拟合采样-演奏法自动标记的数据生成范式，解决了基于语言模型的少样本乐器生成数据短缺问题。在使用生成数据对语言模型进行全量迁移训练的基础上，使用少量真实数据对语言模型和编码器模型进行联合局部微调，以提升音频生成保真度。该项目为日本国立信息学研究所交换期间的研究课题。

角色: 独立项目。 **发表:** [C9]

[软硬件协同设计] 基于 Sketch 的模型参数压缩与适配算子融合

2025 年 5 月 – 2026 年 2 月

内容: 本项目使用专用 Sketch 算法压缩大语言模型权重，并开发解压缩计算融合算子以降低额外开销。我们首先使用针对大模型权重分布特征的 Sketch 算法对模型权重进行压缩，在保证性能的同时突破了 1 比特压缩率限制。其次，使用重要性度量分配 Sketch 压缩空间，进一步减少压缩带来的性能损失。为减少解压缩带来的推理开销和训练性能损失，我们将 Sketch 中的顺序哈希操作映射为等价矩阵乘操作，开发压缩解压缩操作和后续矩阵乘操作融合算子，有效降低额外开销。

角色: 独立项目。 **发表:** [C7]

[编译器] Sketch 加速器生成与优化框架

2024 年 2 月 – 至今

内容: 本项目聚焦于适配多种 Sketch 算法与不同后端硬件的加速器生成优化框架设计。首先，我们定义了可扩展的领域定制语言，支持表示层次化、多 Sketch 等应用场景和算法设计。其次，提出了高效分析性模型预估所需的硬件资源、运行性能、Sketch 预估准确率等关键指标，为后续架构探索提供指引。项目涵盖 FPGA、GPU、CPU、等多种后端，结合实际编译结果与分析性模型预估指标进行架构探索，以优化设计关注的吞吐量或时延指标。项目将与不同 Sketch 算法系统级应用如网络监测、冷热页检测、工作

集预估等应用结合，优化整体运行效率。项目后续工作预计投稿至 ASPLOS 2027。

角色：独立项目。发表：[C6]

[设计自动化] **FPGA 布线并行加速** 2023 年 10 月 – 2024 年 3 月

内容：本项目聚焦于使用粗细粒度结合的并行方式大幅提升 FPGA 布线搜索速度。一方面，本项目使用信号间并行，递归划定并行区域，利用有向无环图解决了信号间潜在冲突问题。另一方面，我们使用源节点到终节点搜索过程的双向并行，结合信号间并行进一步提升了搜索效率。

角色：主要贡献者 & 代码维护者。发表：[C5]

[设计自动化] **逻辑精确综合流程加速** 2023 年 1 月 – 2025 年 1 月

内容：本项目聚焦于精确综合的运行时间的减少和可解问题规模的增加。精确综合依赖 SAT 后端生成最优电路，导致运行时间长且不可预估，且存在可解问题规模小造成后续工艺优化空间有限的问题。观察到不同布尔电路适合不同编码方式之后，本项目利用机器学习方法布尔函数特征预测适合编码以提升 SAT 求解效率。本项目还利用部分最优电路具有最优子结构的特点，设计经验性辅助函数抽取待解问题子问题，并利用子问题设计基于拓扑的和基于子电路的两种增量式 SAT 求解方式。后续将工作开发问题特定 SAT 求解器，预计投稿至 TCAD。

角色：独立项目。发表：[C1, C2]

[设计自动化] **强化学习辅助的逻辑综合功耗优化** 2021 年 7 月 – 2023 年 7 月

内容：本项目使用强化学习建模逻辑综合优化命令搜索问题，通过引入高效功耗建模作为强化学习奖励函数，自动优化设计的功耗-面积-性能帕累托最优前沿。本项目首先通过考虑了毛刺和动态信号的功耗建模方法，利用 SMT 表示扩充了精确综合优化目标，生成最优功耗局部电路，并将其融入传统逻辑优化命令。其次，本项目手动设计电路表示特征作为强化学习环境，以功耗-面积-性能为优化目标训练命令决策模型，生成了特定电路的优化路径，在不影响面积和性能的情况下有效提升了逻辑综合的功耗表现。

角色：独立项目。发表：[J1, C3]

荣誉奖项

• 北京大学三好学生	2024 年
• 北京大学比亚迪奖学金	2024 年
• International Symposium on FPGA 布线比赛第三名	2024 年
• 集成电路 EDA 设计精英挑战赛三等奖	2022 年
• 北京大学本科生科研训练校长基金（理科）	2021 年
• 北京大学新生奖学金	2018 年
• 北京大学信息科学技术学院新生院长奖学金	2018 年

服务

• 课程助教：并行与分布式计算导论，北京大学	2022 年，2023 年
• 官方审稿：IEEE TCAD	

技能

• 编程：C/C++, Python(PyTorch), FPGA HLS, Verilog, CUDA
• 工具：LATEX, Git, Docker/Apptainer
• 语言：中文、英语、日语
• 前北京大学学生交响乐团成员

Sunan Zou

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Personal Profile

I am a fourth-year Ph.D. candidate in the School of Computer Science at Peking University and a member of the PKU-DASYS Lab at the Center for Energy-Efficient Computing and Applications (CECA), advised by Prof. Guojie Luo. I received my B.S. in Computer Science and Technology from Peking University in 2022. My research interests include LLM systems and infrastructure, domain-specific accelerator compilation frameworks, and AI compilers. I currently focus on kernel-schedule co-design for emerging model architectures, as well as model compression and compression-aware operator/architecture design. I also have extensive experience in electronic design automation, particularly logic synthesis optimization. My doctoral research centers on system-level applications of Sketch algorithms, hardware optimization, and automated accelerator generation.

Education

Ph.D. in Computer Architecture

School of Computer Science, Peking University
Advisor: Prof. Guojie Luo

Sep 2022 – Jun 2027 (expected)

Beijing, China

B.S. in Computer Science and Technology

School of Electronics Engineering and Computer Science (EECS), Peking University

Sep 2018 – Jul 2022

Beijing, China

Research Experience

Dots, Rednote

Research Intern, Ace Talent Program.

Topics: xHC Fusion and Scheduling for LLM Training Systems

May 2026 - Now

Beijing, China

Yamagishi Lab, National Institute of Informatics (Japan)

Research Intern, advised by Prof. Junichi Yamagishi.

Topics: Neural-Language Model for Violin Synthesis

Oct 2025 - Apr 2026

Tokyo, Japan

Electrical and Computer Engineering, Carnegie Mellon University

Research Intern, advised by Prof. Larry Pileggi.

Topics: Graph Neural Network-Based Logic Locking Detection

Mar 2021 - Sep 2021

Remote

DASYS Lab, Peking University

Research Intern, advised by Prof. Guojie Luo.

Topics: Reinforcement Learning-Based Logic Optimization

Jul 2019 - Jan 2022

Beijing, China

Publications

- [C9] **Sunan Zou**, Zhe Zhang, Yigitcan Özer, Guojie Luo, and Junichi Yamagishi. “VioLM: A Neural Language Model for Violin Synthesis with Articulation”. *The 27th International Society for Music Information Retrieval Conference (ISMIR)*, 2026. (To Appear)
- [C8] Xiao Tan, Chenyue Li, **Sunan Zou**, and Guojie Luo. “RTLPilot: Skill-Driven Multi-Agent System for Repository-Level RTL Code Migration”. *2026 International Symposium of EDA (ISED)*, 2026. (To Appear)
- [C7] **Sunan Zou**, Xueting Sun, Ziyun Zhang and Guojie Luo. “UltraSketchLLM: Sub-1-Bit LLM Compression via Sketch and Hardware-Friendly Operators”. *63st Design Automation Conference (DAC)*, 2026. (To Appear)
- [C6] **Sunan Zou**, Bizhao Shi, Ziyun Zhang and Guojie Luo. “MuSA: Multi-Sketch Accelerator with Hybrid Parallelism and Coalesced Memory Organization”. *IEEE 42nd International Conference on Computer Design (ICCD)*, 2024.

- [C5] Xinming Wei, Ziyun Zhang, **Sunan Zou**, Kaiwen Sun, Jiahao Zhang, Jiaxi Zhang, Ping Fan, and Guojie Luo. "AceRoute: Adaptive Compute-Efficient FPGA Routing with Pluggable Intra-Connection Bidirectional Exploration". *43th IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2024.
- [C4] Bizhao Shi, Tuo Dai, **Sunan Zou**, Xinming Wei, and Guojie Luo. "ImageMap: Enabling Efficient Mapping from Image Processing DSL to CGRA". *30th International European Conference on Parallel and Distributed Computing (Euro-Par)*, 2024.
- [C3] **Sunan Zou**, and Guojie Luo. "PONO: Power Optimization with Near Optimal SMT-based Sub-circuit Generation". *61st Design Automation Conference (DAC)*, 2024.
- [C2] **Sunan Zou**, Jiaxi Zhang, and Guojie Luo. "Incremental SAT-based Exact Synthesis", *34th Great Lakes Symposium on VLSI (GLSVLSI)* 2024. **[Best Paper Reward]**
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- [J2] Lei Chen, Yiqi Chen, Zhufei Chu, Wenji Fang, Tsung-Yi Ho, et al, and **Sunan Zou (Alphabetical Order)**. "Large circuit models: opportunities and challenges". *Science China Information Sciences Volume 67*, 2024.
- [J1] **Sunan Zou**, Jiaxi Zhang, Bizhao Shi, and Guojie Luo. "PowerSyn: A logic synthesis framework with early power optimization", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems Volume: 43 (TCAD)*, 2023.

Representative Projects

[LLM Training Systems] xHC Fusion & Communication-Aware Scheduling

May 2026 - Now

Contents: Optimized the fine-grained computation and multi-stream state movement introduced by xHC in LLM training within Megatron Core. Designed GPU kernels tailored to xHC tensor shapes and memory-access patterns, and selected fusion boundaries based on data dependencies, data reuse, and GPU execution characteristics. Co-designed Triton forward/backward implementations and custom Autograd paths so activation saving, recomputation, and sparse state materialization align with the fusion boundaries. At the scheduling level, explored shared control computation across sublayers and state management under CUDA Graph and combined 1F1B execution; further explored pre-communicating inactive streams across PP/VPP boundaries to overlap communication with active-path computation. Validated correctness and performance through numerical checks, profiling, and end-to-end training.

Role: Main contributor.

[LLM Algorithm] Violin Audio Model

Oct 2025 - Apr 2026

Contents: This project focuses on the design of a two-stage transfer training framework and data generation paradigm for musical instrument audio generation. First, the project proposes a score tokenization method that accounts for playing techniques, upon which a data generation paradigm is constructed comprising score legalization, dynamic fitting sampling, and automatic articulation annotation, addressing the data scarcity problem. Building on full-parameter transfer training of the language model using generated data, a small amount of real data is employed for joint partial fine-tuning of both the language model and the encoder model to improve audio generation fidelity. This project was conducted as a research topic during an exchange visit at the National Institute of Informatics (NII), Japan, with a planned submission to ACM Multimedia 2026.

Role: Independent project.

Publication: [C9]

[SW/HW Co-design] Sketch-Based LLM Compression & Hardware-Friendly Kernel

May 2025 - Feb 2026

Contents: This project applies Sketch algorithms to compress LLM weights and develops fused compression and decompression kernels to reduce associated overhead. We first compress model weights using a Sketch algorithm tailored for LLMs, achieving sub-1-bit compression rates while maintaining model performance. To reduce the inference overhead and training performance loss introduced by decompression, we reformulate the sequential hashing operations in Sketch as equivalent matrix multiplication operations, and develop fused kernels that integrate compression/decompression with computation, effectively reducing additional overhead.

Role: Independent project.

Publication: [C7]

[Compiler] Sketch Accelerator Design & Auto-Generation**Feb 2024 - Now**

Contents: This project focuses on the design of an accelerator generation and optimization framework adaptable to multiple Sketch algorithms and diverse backend hardware targets. First, we define an extensible domain-specific language (DSL) capable of expressing hierarchical and multi-Sketch application scenarios and algorithm designs. Second, we propose efficient analytical models to estimate key metrics, providing guidance for subsequent architecture DSE. The project spans multiple backend targets including FPGAs, GPUs, and CPUs. The framework will be integrated with system-level applications of various Sketch algorithms, including network monitoring, hot-cold page detection, and working set estimation, to optimize overall runtime efficiency. Follow-up work from this project is planned for submission to ASPLOS 2027.

Role: Independent project. **Publication:** [C6]

[Design Automation] Multi-Level Parallel FPGA Routing**Oct 2023 - Mar 2024**

Contents: This project focuses on significantly accelerating FPGA routing search through a combined coarse- and fine-grained parallelization approach. On one hand, the project employs inter-signal parallelism by recursively partitioning parallel regions and leveraging directed acyclic graphs (DAGs) to resolve potential inter-signal conflicts. On the other hand, we introduce bidirectional parallelism in the search process from source nodes to sink nodes, which, combined with inter-signal parallelism, further improves overall search efficiency.

Role: Main contributor & Code maintainer. **Publication:** [C5]

[Design Automation] SAT-Baed Incremental Synthesis Acceleration**Jan 2023 - Jan 2025**

Contents: This project focuses on reducing the runtime and increasing the solvable problem scale of exact synthesis. Upon observing that different Boolean circuits are better suited to different encoding schemes, this project employs machine learning methods to predict appropriate encodings from Boolean function features, thereby improving SAT solving efficiency. The project further exploits the optimal substructure property present in certain optimal circuits, designing heuristic auxiliary functions to extract subproblems from the target problem, and developing two incremental SAT solving strategies based on topology and a sub-circuit. Follow-up work will develop a problem-specific SAT solver, with a planned submission to TCAD.

Role: Independent project. **Publication:** [C1, C2]

[Design Automation] Reinforcement Learning for Logic Optimization**Jul 2021 - Jul 2023**

Contents: This project formulates the logic synthesis optimization command search problem as a reinforcement learning task, and optimizes circuit designs by incorporating an efficient power modeling method as the reinforcement learning reward function. First, the project extends the exact synthesis optimization objective using SMT formulations with a power modeling approach that accounts for glitches and dynamic signals, generating locally optimal low-power circuits and integrating them into conventional logic optimization commands. Second, the project manually engineers circuit representation features as the reinforcement learning environment, trains a command decision model with power, area, and performance as joint optimization objectives, and produces circuit-specific optimization sequences that effectively improve the power characteristics of logic synthesis without compromising area or performance.

Role: Independent project. **Publication:** [J1, C3]

Honors and Awards

- | | |
|---|------|
| • Honors for Merit Student (三好学生), Peking University | 2024 |
| • BYD Scholarship (北京大学比亚迪奖学金), Peking University | 2024 |
| • ISFPGA FPGA Routing Contest Third Place, ISFPGA Committee | 2024 |
| • Principal's Fund for Undergraduate Research Training (Natural Science), Peking University | 2021 |
| • Freshman Scholarship (新生奖学金), Peking University | 2018 |
| • Freshman Scholarship (院长新生奖学金), School of EECS, Peking University | 2018 |

Services

- **Official Reviewer:** TCAD

- **Teaching Assistant:** Introduction to Parallel and Distributed Computing, Peking University 2022, 2023

Skills

- **Programming Languages:** C/C++, Python(Pytorch), FPGA HLS, Verilog, CUDA
- **Tools:** LATEX, Git, Docker/Apptainer
- **Languages:** Chinese, English, and Japanese.
- Former Member of Peking University Student Orchestra